STAFF REVIEW · H446 143 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Router.

2. AND.

3. Holding a CPU's working data/ALU results.

4. Binary search: O(log n) rather than O(n).

I Do — source worked steps

1. HTTP application data — Identify the relevant information and make a prediction.

2. TCP segment in IP packet — Explain the change using correct technical terms.

3. Ethernet frame — HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

We Do — guided CHECK

1. Incoming Ethernet frame
Identify the relevant information and make a prediction.
Route IP packet
Explain the change using correct technical terms.
Outgoing Wi-Fi frame
Router removes the incoming link-layer framing, processes/forwards the IP packet and builds new outgoing link-layer framing. IP hop limit/TTL normally decreases; end-to-end transport data need not change.

You Do — source mark schemes

1. Application user-facing network protocols [2], transport end-to-end delivery/ports [2], internet addressing/routing [2], link local frame transport [2].

2. Application message [1], TCP header/segment [2], IP header/packet [2], frame encapsulation [1].

3. Remove old link frame [2], inspect IP destination/routing [2], reduce TTL/hop limit and update necessary header fields [2], encapsulate with new link frame [2].

Exit Ticket — source answer

1. False. Frames are recreated for each local link/hop.